

Saurabh Yadav

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Research Interests: Computer Vision and Deep Learning

EDUCATION

IIIT Delhi <i>PhD in Computer Science and Engineering (graduating soon)</i>	New Delhi, India <i>Jan. 2021 – present</i>
University of Delhi <i>M.Sc. in Computer Science</i>	New Delhi, India <i>July 2018 – May 2020</i>
HansRaj College, University of Delhi <i>B.Sc.(Hons) in Computer Science</i>	New Delhi, India <i>July 2015 – May 2018</i>

EXPERIENCE

Project Associate <i>IIT Kanpur</i> Working on continual domain adaptation for audio-visual event localization problem	Nov. 2025 – present
Research Intern – AI and Machine Learning <i>GE Aerospace Research</i> - Designed a reasoning-based vision-language model to extract insights from unstructured multi-modal data using Large Language Models. - Developed and deployed a full-stack AI-powered decision support tool for aerospace engineers using FastAPI and Vue.js.	Mar. 2025 – Aug. 2025 <i>Bangalore, India</i>
Research Intern – Computer Vision <i>Sony Research India</i> - Engineered a multilingual lip-syncing system using diffusion models and deep learning for real-time video translation between English and Hindi. - Achieved significant performance improvements through customized loss functions and temporal consistency.	Feb. 2024 – July 2024 <i>Remote</i>

PUBLICATIONS

Published

- S. Yadav** and K. R. Jerripothula, “FCCNs: Fully complex-valued convolutional networks using complex-valued color model and loss function”, ICCV 2023 [[pdf](#)][[CORE A*](#)].
- S. Yadav**, A. Gupta, and K. R. Jerripothula, “SAMwave: Adapting Segment Anything Model to Difficult Tasks”, BMVC 2025 (oral) [[pdf](#)][[CORE A](#)]

In Submission

- S. Yadav** and K. R. Jerripothula, “DCS²Net: Deep Complex Spatio-Spectral Networks With Complex Visual Inputs,” CVPR 2026 [[pdf](#)].
- A. Gupta, **S. Yadav**, and K. R. Jerripothula, “Adaptive knowledge transfer for class-incremental semantic segmentation,” ICLR 2026 [[pdf](#)].

PROJECTS

Complex-Valued Neural Networks for Vision applications

- Built complex-valued color models to convert real-valued RGB images to complex-valued images.
- Built complex-valued CNNs and token-based architectures for image classification and segmentation.
- Impact:** Demonstrated superior performance over real-valued counterparts by capturing complex-valued representation.

Effective adaptation of segment anything models for complex image and video segmentation

- Designed a novel adapter-based method for fine-tuning foundation models on complex segmentation tasks.
- Enhanced performance of segment anything models on camouflaged images, Polyp images, shadow images, etc.
- Impact:** Demonstrated parameter-efficient fine-tuning of foundational model using real and complex wavelet transform.

Image-Conditioned Diffusion Models for personalized image generation

- Developed a hybrid T+I2I diffusion model based on SD-XL for prompt-based image synthesis.
- Implemented dynamic latent editing for style and color control via textual input.
- **Impact:** Enabled high-fidelity, personalized image generation with user control over color via textual prompts.

Continual Semantic Segmentation

- Proposed a class-incremental segmentation method using knowledge distillation and regularization.
- **Impact:** Successfully mitigated catastrophic forgetting, enabling models to learn new semantic classes sequentially with minimal performance drop on previous tasks.

AWARDS AND ACHIEVEMENTS

Doctoral Consortium

Nov. 2025

- Selected for presenting research work at BMVC 2025.

Google Travel Grant

Sept. 2023

- Awarded full funding to present research at ICCV 2023.

ICCV Diversity, Equity and Inclusion Award

Aug. 2023

- Selected among 147 recipients globally for promoting diversity in AI research.

UGC-NET JRF Fellowship

Jan. 2020

- Top 0.5% among national candidates qualified for Junior Research Fellowship in Computer Science.

TEACHING AND OTHERS

Teaching Services

- Teaching Assistant, Introduction to Mathematical Logic (*Feb-May 2021*)
- Head Teaching Assistant, Computer Vision (*Jan-May 2024*)
- Head Teaching Assistant, Computer Vision (*Jan-May 2025*)

Talks

- Hosted talk on Machine Learning Model Optimization by Weights and Biases (Wandb).
- Presented research work on Autoencoders at CyFuse, IIIT Delhi
- Presented work on Complex-valued networks at VIMS lab, IIT Kanpur.
- Gave a guest lecture at the Introduction to Intelligent Systems course, IIIT Delhi.

Peer Review Services

- Member of Program Committee, PyData Global 2022
- Reviewer for WACV 2024, NeurIPS 2025, ICCV 2025, IJCNN 2025, ICASSP 2026

TECHNICAL SKILLS

Programming Languages: Python, C++, JavaScript, HTML, CSS, Lua

Machine Learning Frameworks: PyTorch, TensorFlow, Keras, FastAI

Libraries and Tools: NumPy, SciPy, OpenCV, Matplotlib, Git, Vim, VS Code, QMK Firmware, ZMK Firmware

Areas of Expertise: Computer Vision, Deep Learning, NLP, Semantic Segmentation, Vision-Language Models, Diffusion Models, Knowledge Transfer